

EMERGING MATERIALS, RARE EARTHS AND CRITICAL MINERALS

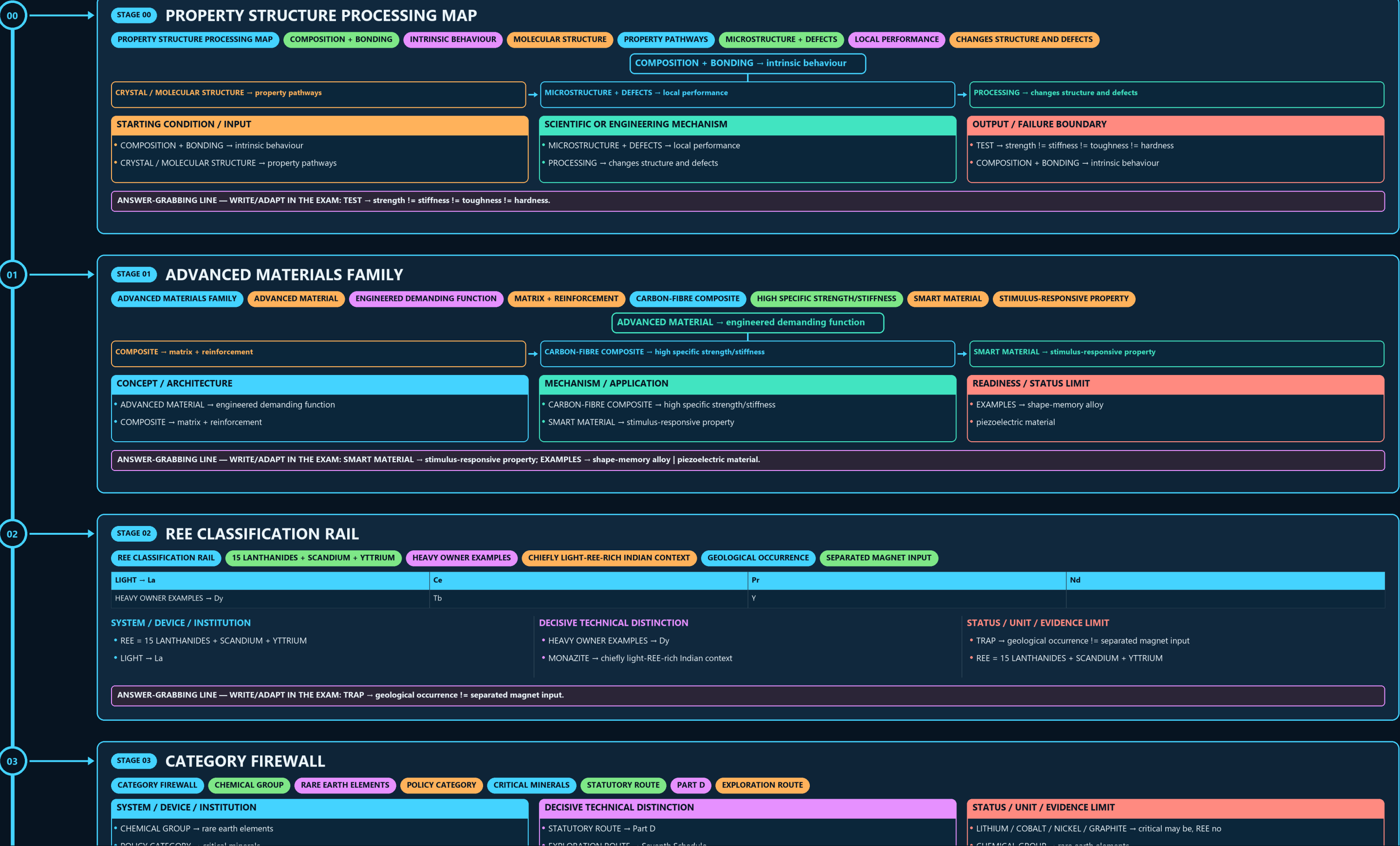
PROPERTY STRUCTURE PROCESSING MAP → ADVANCED MATERIALS FAMILY → REE CLASSIFICATION RAIL → CATEGORY FIREWALL → PERMANENT MAGNET CAPABILITY LADDER → BATTERY MINERAL ROLE MATRIX

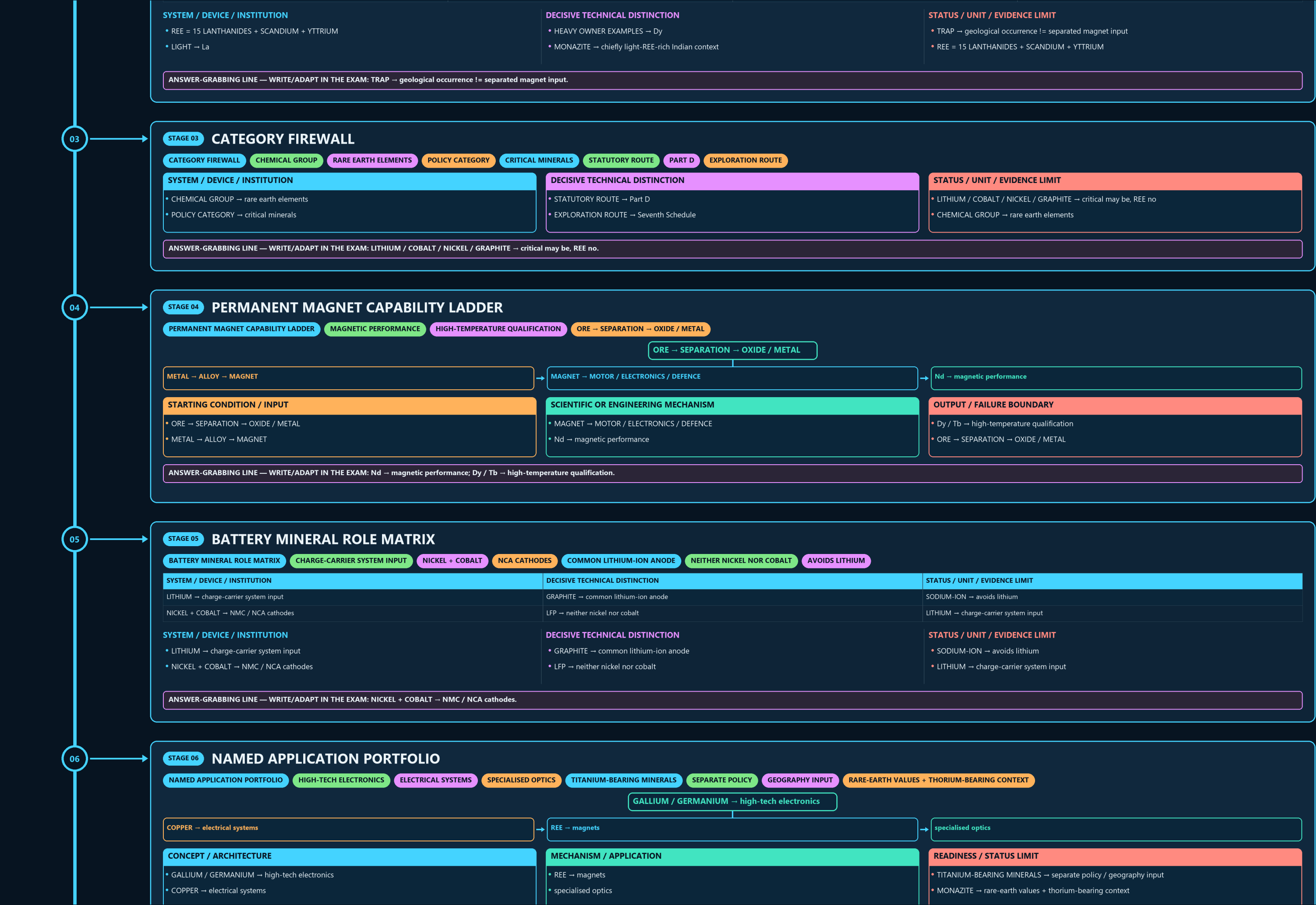
Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • active generation g1 • explicit approval of this exact generation is required • all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles





ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: NICKEL + COBALT → NMC / NCA cathodes.

06

STAGE 06 NAMED APPLICATION PORTFOLIO

NAMED APPLICATION PORTFOLIO HIGH-TECH ELECTRONICS ELECTRICAL SYSTEMS SPECIALISED OPTICS TITANIUM-BEARING MINERALS SEPARATE POLICY GEOGRAPHY INPUT RARE-EARTH VALUES + THORIUM-BEARING CONTEXT

GALLIUM / GERMANIUM → high-tech electronics

COPPER → electrical systems

REE → magnets

specialised optics

CONCEPT / ARCHITECTURE

- GALLIUM / GERMANIUM → high-tech electronics
- COPPER → electrical systems

MECHANISM / APPLICATION

- REE → magnets
- specialised optics

READINESS / STATUS LIMIT

- TITANIUM-BEARING MINERALS → separate policy / geography input
- MONAZITE → rare-earth values + thorium-bearing context

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TITANIUM-BEARING MINERALS → separate policy / geography input.

07

STAGE 07 MINE TO TECHNOLOGY VALUE CHAIN

MINE TO TECHNOLOGY VALUE CHAIN RECOVERED FEED WITH PURITY/ECONOMIC LIMITS EXPLORE → ASSESS → DEVELOP → EXTRACT BENEFICIATE → SEPARATE / REFINE

EXPLORE → ASSESS → DEVELOP → EXTRACT

BENEFICIATE → SEPARATE / REFINE

OXIDE / METAL / ALLOY → COMPONENT

COMPONENT → BATTERY / MAGNET / CHIP / SYSTEM

RECYCLE → recovered feed with purity/economic limits

STARTING CONDITION / INPUT

- EXPLORE → ASSESS → DEVELOP → EXTRACT
- BENEFICIATE → SEPARATE / REFINE

SCIENTIFIC OR ENGINEERING MECHANISM

- OXIDE / METAL / ALLOY → COMPONENT
- COMPONENT → BATTERY / MAGNET / CHIP / SYSTEM

OUTPUT / FAILURE BOUNDARY

- RECYCLE → recovered feed with purity/economic limits
- EXPLORE → ASSESS → DEVELOP → EXTRACT

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: COMPONENT → BATTERY / MAGNET / CHIP / SYSTEM.

08

STAGE 08 DEPENDENCY HEAT MAP

DEPENDENCY HEAT MAP SECURE MINE ALONE IS PARTIAL SECURITY UPSTREAM → GEOLOGY MIDSTREAM → SEPARATION

UPSTREAM → geology

MIDSTREAM → separation

DOWNSTREAM → cell

ENABLERS → technology

CONCEPT / ARCHITECTURE

- UPSTREAM → geology
- MIDSTREAM → separation

MECHANISM / APPLICATION

- DOWNSTREAM → cell
- ENABLERS → technology

READINESS / STATUS LIMIT

- VERDICT → secure mine alone is partial security
- UPSTREAM → geology

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VERDICT → secure mine alone is partial security.

09

STAGE 09 CIRCULAR MATERIALS LOOP

CIRCULAR MATERIALS LOOP RE-ENTER MANUFACTURE SUBSTITUTE + EFFICIENCY REDUCE PRIMARY-MATERIAL PRESSURE

DESIGN → durability

USE → reuse / remanufacture

COLLECT → sort → pre-process

RECOVER → purify → re-enter manufacture

SUBSTITUTE + EFFICIENCY → reduce primary-material pressure

CONCEPT / ARCHITECTURE

- DESIGN → durability
- USE → reuse / remanufacture

MECHANISM / APPLICATION

- COLLECT → sort → pre-process
- RECOVER → purify → re-enter manufacture

READINESS / STATUS LIMIT

- SUBSTITUTE + EFFICIENCY → reduce primary-material pressure
- DESIGN → durability

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RECOVER → purify → re-enter manufacture; SUBSTITUTE + EFFICIENCY → reduce primary-material pressure.

09

STAGE 09 CIRCULAR MATERIALS LOOP

CIRCULAR MATERIALS LOOP RE-ENTER MANUFACTURE SUBSTITUTE + EFFICIENCY REDUCE PRIMARY-MATERIAL PRESSURE



CONCEPT / ARCHITECTURE

- DESIGN → durability
- USE → reuse / remanufacture

MECHANISM / APPLICATION

- COLLECT → sort → pre-process
- RECOVER → purify → re-enter manufacture

READINESS / STATUS LIMIT

- SUBSTITUTE + EFFICIENCY → reduce primary-material pressure
- DESIGN → durability

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RECOVER → purify → re-enter manufacture; SUBSTITUTE + EFFICIENCY → reduce primary-material pressure.

10

STAGE 10 MINE-LIFE SAFEGUARD CHAIN

MINE-LIFE SAFEGUARD CHAIN CUMULATIVE IMPACT + CONSULTATION TAILINGS CONTROLS FRA DUE DILIGENCE STABLE DUMPS NATIVE RESTORATION FINANCE + MONITORING; PLAN ACHIEVED OUTCOME

OPERATIONS → dust	noise	water	tailings controls
COMMUNITY → FRA due diligence	disclosure	grievance	
RECLAMATION → topsoil	stable dumps	native restoration	

STARTING CONDITION / INPUT

- SITING → cumulative impact + consultation
- OPERATIONS → dust
- tailings controls

SCIENTIFIC OR ENGINEERING MECHANISM

- COMMUNITY → FRA due diligence
- RECLAMATION → topsoil
- stable dumps

OUTPUT / FAILURE BOUNDARY

- native restoration
- CLOSURE → finance + monitoring; plan != achieved outcome

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSURE → finance + monitoring; plan != achieved outcome.

11

STAGE 11 INDIA INSTITUTIONS STATUS AND PYQ RAIL

INDIA INSTITUTIONS STATUS AND PYQ RAIL STATE GRANT + S.11D CENTRAL AUCTION DISTINCT ROLES MONAZITE ROUTE OVERSEAS ROUTE DEEP-OCEAN EXPLORATION BOUNDARY



CONCEPT / ARCHITECTURE

- MINES → policy
- GSI → survey/explore
- IBM → plans/conservation

MECHANISM / APPLICATION

- STATE GRANT + s.11D CENTRAL AUCTION → distinct roles
- IREL / DAE → monazite route
- KABIL → overseas route

READINESS / STATUS LIMIT

- MoES / NIOT / ISA → deep-ocean exploration boundary
- REPORT → LAW → APPROVAL → MOU → EXPLORATION → PRODUCTION → OUTCOME

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: REPORT → LAW → APPROVAL → MOU → EXPLORATION → PRODUCTION → OUTCOME.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT THE KEY ADVANCED DISTINCTION IS BETWEEN GEOLOGICAL PRESENCE AND A COUNTRY MAY HAVE RESOURCES BUT STILL REMAIN VULNERABLE THEREFORE, CRITICAL-MINERAL POLICY IS SIMULTANEOUSLY A MINING POLICY, INDUSTRIAL MINISTRY OF MINES
POLICY AND MISSION-LEVEL ANCHOR. THE NATIONAL GEOSCIENTIFIC SURVEY AND EXPLORATION AGENCY — NOT MINING-PAN APPROVAL, SCIENTIFIC MINING AND MINERAL CONSERVATION

OPTIONAL SCIENTIFIC / ENGINEERING DEPTH

- CAUTION: The key advanced distinction is between geological presence and usable strategic capability .
- CAUTION: A country may have resources but still remain vulnerable if refining and component manufacturing are concentrated elsewhere.
- CAUTION: Therefore, critical-mineral policy is simultaneously a mining policy, industrial policy, trade policy and strategic policy.
- Ministry of Mines: policy and mission-level anchor.

READINESS, UNIT OR STATUS LIMIT

- GSI: the national geoscientific survey and exploration agency — not the mine developer, lease-granting authority or sectoral regulator.
- IBM: mining-plan approval, scientific mining and mineral conservation
- State governments grant concessions, with the Centre auctioning Part D minerals under s.11D .
- IREL (India) Ltd (DAE): monazite processing for rare earths and thorium.

COMPARATIVE TECHNOLOGY / GOVERNANCE NUANCE

- KABIL: overseas sourcing/exploration instrument.
- MoES / NIOT — Deep Ocean Mission: seabed polymetallic-nodule exploration under an ISA contract.
- Mineral Security Partnership: plurilateral investment-coordination platform India has joined.
- NCMM: mission framework covering exploration, mining, processing, recycling and stockpiles.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.